TABLE OF CONTENTS

1.- OBJECT AND SCOPE

2.- GENERAL STANDARDS

3.- LIST OF DOCUMENTS

4.- DESIGN VERIFICATION

4.1. Protocol

4.2. Activity 36 – Piping design

a) Plot plans (Items 3623/3630)
b) PFD´s and P&ID´s (Items 3621/3622)
c) Key plan (Item 3631)
d) Drawings of piping plan views (Item 3624/3650)
Piping studies for physical model (Item 3651)
Rack drawings (Items 3626/3652)
Piping drawings for disassembly (Item 3629)
Draft drawings from 3D (Item 3627)
e) Auxiliary drawings, vents, drains (Item 3653)
f) Isometrics (Items 3660/3661)
Tie-in´s (Item 3662)
g) Ducts (Item 3663)
h) Piping take-offs (Item 3665)
i) Material list (Item 3666)
j) Physical model assembly (Item 3670)
k) Package units (Item 3668)
l) Piping design in 3D (Item 3640)
m) 3D administration (Item 3696)
n) 2D administration (Item 3697)

5.- VERIFICATION OF PIPING MATERIALS

5.1. Activity 34 – Material specifications and MTO

a) Specifications of piping material (Item 3410)
b) Material Take-off (MTO's) (Item 3420)
c) Isometrics, tie-in´s and its material lists (Item 3440/3445)
d) Material requirements (RM's), technical comparisons and vendor documentation (Item 3450)
e) Special elements of pipes (H Items) (Item 3430)
f) Fluid List and Pipe List (Item 3435)
g) European Directive of pressure equipments (Item 3460)
h) Calculations of thickness to internal and external pressure (Item 3470)
i) Package Units (Item 3497)

6- VERIFICATION OF PIPING STRESS AND SUPPORTS

6.1. Activity 35 – Stress and supports

a) Calculation of stresses in pipes and equipments (Item 3510)
b) Ducts (Item 3511)
c) Calculation and design of supports (Item 3520)
d) Support List (Item 3530)
e) Special elements (Item 3540)
f) European Directive of pressure equipments (Item 3560)
g) Standard supports in 3D (Item 3596)
h Package Units (Item 3594)
i) Material requirements (RM's), technical comparisons and vendor documentation (Item 3540)
j) Material Take-off of tubular material (MTO's) (Item 3530)

Annexed 1.Verification plan

1.- OBJECT AND SCOPE

This Job Instruction defines the general standards and the detailed procedure for the verification of documents produced by the Piping Department.

This Job Instruction is part of the **Intecsa Industrial** Quality System and it has been prepared taking into account the ISO-9001 Code.

2.- GENERAL RULES

All of the verification task will be performed in accordance with the indications of the **"Procedure of Document Verification" (PR-34)**.

The verification levels will be collected in the Department Verification Plan, except in the Projects in which variations to it are defined.

As a general rule, the verification of documents will take place by a person other than the author of them, although, depending on the document type and of the level of development established for the Job Order in question, the verification of some documents may take place by the author of the same. In this case the author must also respect this Job Instruction as if it were a different person. See Verification Plan in Annexed I.

The document verifier will check that his own comments or the other department comments, have been properly included before signing the document.

The content of the procedures which make up the **Manual of Procedures for the Execution of Intecsa Industrial Projects,** must be taken into account during the process of verification, especially in the case of drawings, the **"Procedure of Execution of Drawings in PC" (PR-38)** and the **"Procedure of Drafting" (PR-37)** and in all cases the **Job Instructions** and the standard drawings of the Department which are applicable, as well as the specifications of the Project itself.

3.- LIST OF DOCUMENTS

The documents which must be consulted in each case in order to perform the verification tasks are given in the verification lists (paragraph 4) with the following numbering:

1. Project basic data, received with initial documentation and with Basic engineering, or defined in the first stage of work.
2. Material specifications.
3. Engineering Diagrams (P&ID)
4. Preliminary studies.
5. Plot plans.
6. Key plan.
7. Piping plan view drawings.
8. Piping studies for model.
9. Auxiliary drawings, vents and drains.
10. Isometrics and Tie-in´s
11. Civil Work Drawings.
12. Drawings of Mechanical Equipments (rotary, vessels, exchangers, etc.).
13. Electricity Drawings (trays, area classification, etc.).
14. Instruments Drawings (local cabinets, panels, etc.).
15. Analysis of Stresses and Supports.
16. Dimensional Standards.
17. Piping List.
18. Equipment List.
19. Drawing List.
20. Fluid List.
21. MTO Sheets.
22. Index of classes of Material Specifications.
23. Item “H” List.
24. Material Requirements.
25. Offers Technical Comparison.
26. Suppliers Documentation.
27. 3D Electronic Model.
28. Duct Drawings.
29. Supports List (special and standard).
30. Stress and Support Isometrics.

When the numbering is shown in brackets, is applied only in the case of applying the project.

4.- DESIGN VERIFICATION

4.1. Protocol

The general protocol consists of verifying aspects, applicable to any document.

Specific protocols relate the particular aspects to verify of each document.

- **Format.**

  Check if the used format corresponds to the specified for the project between those standardized in **Intecsa Industrial** or, where appropriate, the required by the customer.

  In drawings produced at the computer it will be check that all file components (layers, colours, thicknesses, cells, etc.) are correct and comply with the **Procedure in executing drawings PC (PR-38)**. The Discipline Leader (DL) will verify the compliance of this point before issue the drawing.

- **General orientation of the drawing and its details regarding the North.**

  Check that the drawing contains the North symbol on top, and its direction is correct.

  Check if that is really, the entire plane in general and all the details are correctly located in respect of the North. Special care should be taken to check the situation of points of reference, pillars and foundations.

- **Limits of representation and delivery**

  Representation drawing limits will be checked in consistency with the realized distribution on the overall plant or key plan corresponding.

  Battery limits and supply that might exist will be checked and the existing connections and facilities, (Tie-in's).

- **Location coordinates of all drawing elements**

  It will be checked for every of the elements shown, that the reorganization thereof is possible from the coordinates or dimensions shown.

  Not issue any plan that does not meet this requirement, unless a clear indication of "HOLD".

  It will be checked that there are not coordinates and dimensions containing redundant information.

- **Definition of all singular points and its corresponding situation**

  This section refers to layouts, interconnection points, ladders or stairs, step holes, areas of disassembly, etc.

- **Dimensions of all parts**

  It will be checked the dimensioning that of all components contained in the drawing, comply with the Development Level established for the corresponding Work Order, and that the dimensioning has been performed in common units for all drawings of the Project.

  In no case standard parts whose dimensions are determined by dimensional standards or supplier catalogue (elbows, tees, flanges…), will be dimensioned.

- **Elevation of the elements**

  It will be checked that the drawing includes the necessary elevations and that they have been set from the same reference level for all levels of the project and using the abbreviations and symbols standardized.

- **Checking the accuracy of all dimensions**

  This checking will be made taking into account documents that have guided the implementation of the drawing that verified (equipment, structures, ditches, etc.). Clearances shall be checked for lowpass brackets, braces, beams lane, pipe pitch and man pitch, disassembly spaces, access and hallways.

  It will be checked that there is no interference between the buried pipes, cable trenches, ditches and foundations. The design of drainage networks is out of piping range. (By Civil).

  It will be checked that stairs, handrails and platforms referred to standards have no interference with equipments or piping. You have to check the escape ladders in risk areas.

- **Checking plants, sections and details**

  It will verify that details or sections are correctly stated, as the **"Drafting procedure" (Procedure PR-37)** and specific design standards of the Department and are identified with the nomenclature and symbolism in the Project application. It will be checked the consistency in relation between plants, sections and details.

- **Reference Drawings and General Notes**

  It will be verified that all applicable reference drawings are mentioned in a single drawing of the group. It will be checked that in the rest of the drawings there is a reference of that one.

  Likewise, it will act on the General Notes.

- **Letters. Seals. Dates.**

  You should see only notes that, being particular to the drawing in question, are not on the drawing mentioned in the previous section.

  If the drawing is not full edit, it will be checked that have been recirculated all outstanding areas of it, and that are been listed next to the seal all elements affected by the indication of "HOLD" or "PENDING".

  When the drawing must be issued only with a valid part of information, or valid only for certain purposes, check that it has been made clear with the legend "VALID ONLY FOR...", or similar.

  It will be checked that each group of drawings follows a logical sequence enumeration as well as the seal box data, such as signatures, document number and revision.

- **Reviews**

  In each review it will be indicated, by recirculating, the change or changes, marking the corresponding revision (letter or number, as applicable) within a triangle, and indicate in the appropriate box seal the description "Modified the recirculated" or defining directly the change, if this extension allows. It will be checked that there is correspondence between the specified revision in the modified areas and the seal box.

  In the case of a large modification, changes will not be recirculated and in the seal box there will be the description "General Review". This paragraph shall be limited to the maximum to avoid problems on site.

  It will be checked that the previous reviews have been deleted for the recirculated and includes only the editing to issue. On special occasions it is required not to remove previous reviews indications.

- **Ladders**

  It will be checked that in each case the ladder is appropriate to the information level that is transmitted, and that in fact, the drawing corresponds to the indicated scale.

- **Signatures**

  It will be verified that there are planned signatures in the seal box corresponding to the above steps to check, and supplemented with those for the same.

- **Symbology**

  It will be check that the symbols used in all documents to issue, correspond to the specified for the project

4.2. Activity 36 – Piping Design

a) Plot Plans (Items 3423/3630)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• North orientation of the plant, definition of the origin of coordinates, parcel / battery limits situation, access location, elevation and / or pavement.	1
• Minimum safety distances between units, facilities, package units or equipments that require.	1
• Entrances and clearances for assembly, disassembly and maintenance of facilities.	1
• Provision of space for future expansions, if applicable.	1
• Location of internal streets, buildings and structures, piping and cable trays (if applicable) with coordinates and identification of pillars.	11, 13
• Location of curbs, ditches and canals (aerial or buried), pits, etc.	11
• Situation in coordinates and representation of equipment and them operating platforms.	3, 4, 11, 12
• Symbols location for service stations, eyewash showers, etc., (This information must be submitted by facilities).	7, 8
• Check the list of equipments and their elevations.	1, 18, 19

b) PFD´s and P&ID´s (Items 3621/3622)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Delineation of PFD's and P&ID's will be subject to the described in the **"Procedure of Drafting" (PR-37)** and **"Procedure of Execution of Drawings in PC" (PR-38)**. Verification of the data included in these documents is the responsibility of the Process Department.	3

c) Key Plan (Item 3631)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Piping plan drawing limits representation and reference to its number and continuation between them.	7, 19
• At all piping key plans must appear on the seal box, in a visible place the notation "VALID ONLY AS KEY PLAN" to avoid misuse.	6

d) Drawings of piping plan views (Items 3624/3650)
Piping studies for physical model (Item 3651)
Rack drawings (Items 3626/3652)
Piping drawings for disassembly (Item 3651)
Draft drawings from 3D (Item 3627)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• North orientation of the plant, parcel / battery limit situation, elevation and / or pavement.	5, 6, 11
• Location of streets, buildings and structures (coordinates and indication of pillars).	5, 11
• Equipment with connections location and representation, supports (if applicable), foundations and platforms with its elevation.	5, 11, 12
• Checking access and clearances for assembly, disassembly and maintenance of facilities	1, 5
• Representation of pipe networks with all its components and proper installation of them, according to their types. Identification of instruments and other items that require it.	2, 3, 5, 9, 11, 12, 13, 14, 15, 16
• Identification of pipes with a number of line, class, special requirements (type of heat tracing, insulation, etc) and flow direction.	3, 17
• General sizing of different lengths of pipe for proper disposal of them in plan and elevation.	2, 16
• Height of discharge of security valves with respect to the collector of the flare.	
• Points for discharging drainages.	

e) **Auxiliary drawings, Vents, Drains (Item 3653).**

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Definition and identification of different types of typical details by function and installation.	1, 3
• Composition and assembly of elements by class of material specification, particular notes.	2, 3

f) Isometrics (Items 3660/3661)
Tie-ins (Item 3662)

Scope of verification	Documents needed for verification(see section 3 of this IT))
• North orientation. It should be consistent with the north of piping plan drawings.	5, 7 (8)
• Position coordinates or reference to situation of pillars.	5, 7 (8)
• Identification of equipments (axes) and connections.	5, 7 (8), 12
• Line number, class, special requirements (such tracing, isolation, etc.) and flow direction.	3, 17
• Layout and general orientation of the pipeline route as the intended design of pipes in the plant of pipes (or model).	7, (8)
• Representation of all components and proper installation of them, according to their types, and identification of instruments and other items that require it.	3, 7, (8)
• Situation of drains and vents.	7, 9
• General Sizing of individual pipe lengths and instruments, non standard components that require it and joint thickness.	2, 7, (8), 14, 16

Scope of verification	Documents needed for verification(see section 3 of this IT))
• Indication of continuation in line ends and / or branches in other drawings or isometric.	7, 19
• It will be checked that the items displayed in the list of materials match up with the designed elements.	2, 10

g) Ducts (Item 3663)

This scope will be determined for each project according to the scope awarded to the specialty.

h) Piping Measuring (Item 3665)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Check plot plans and diagrams to use (latest revision), specifications and applicable standards.	2, 3, 5
• Verify that have used the established formats in the Work Order and that they comply with the instructions previously received in the measurement that the Materials Specialist has done (header type, code elements and assemblies, etc.).	1, 9
• It will be checked that, in the case of plant drawings, material sheets of each line cover all pipes represented therein, except, if applicable, of those in isometric. Also, check that the measures are of all lines with that appear in the P & Ids.	3, 7, 9, 10, 17, 21

i) Material List (Item 3666)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• List of materials with descriptions, sizes and quantities required.	2

j) Assembly of Physical Model (Item 3670)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• North orientation, area/battery limits with its coordinates.	5, 6, 11
• Location of roads, buildings and structures (coordinates and display of pillars).	5, 11
• Location and installation of equipments, connections, supports or foundations and platforms with its elevation.	5, 11, 12
• Check access and clearances for assembly, disassembly and maintenance of facilities.	1,5
• Identification of pipes with the corresponding coloured material and labels with the line number, class, special requirements (type of heat tracing, insulation, etc.) and flow direction.	3, 17
• Representation of drainage networks, cable trenches, ditches.	11, 13, 14
• Representation of pipe networks with all its components and proper installation, according to their types, identification tags and other instruments that are required.	2, 3, 5, 8, 9, 10, 11, 12, (13), 14, 16

k) Package Units (Item 3668)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Checking general dimensions. • Points of pipe connection (diameter, rating, class, isolation, situation, height, etc.). • Coordinates or datum and area elevations and maintenance walkways. • Disassembly areas of equipments (exchangers, catalysts, etc.). • Classified areas (if appropriate). • Drainages or leaks points. • Heights of safety valves with discharge torch.	1, 2, 3, 5, 11, 12, 13, 14, 17, 18

l) Piping Design in 3D (Item 3640)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Verification of design in the 3D model will be continuously through established tools (Review). The DL. must verify that all changes, errors or additions are included in the 3D model. This verification only will be extended to elements that are within the scope of the piping specialty (including mechanical equipment and on line tools). The 3D design will comply with all the provisions of the preceding points of this IT with regard to key plans, isometric, measurements, plant and rack drawings, etc.	1, 2, 3, 5, 9, 12, 14, 16, 17, 18, 26

m) 3D Administration (Item 3696)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• North orientation of the plant, origin of coordinates, parcel/battery limits situation with its coordinates, location of access, terrain elevation and/or location of roads and pavement.	5, 6, 7, 10, 27
• Division of project in areas and process units.	27
• It will be checked that the classes of materials included in the model agrees closely with those issued by the specialty materials	2, 27
• Verify the main dimensions of the items included in the catalogue needed for design against the documentation of suppliers.	26
• It will be checked that the export of materials for isometric files is used as required, in terms of identifying the line (as line list) and name of the isometric (including number of sheets).	27, 17
• It will be checked that the drawing or isometric format to issue, matches and includes the information required by the project.	

n) 2D Administration (Item 3697)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• North orientation of the plant, origin of coordinates.	2, 7, 10
• It will be checked that the classes of materials included in the model agrees closely with those issued by the specialty materials.	2
• It will be verified the main dimensions of the items included in the catalogue needed for design against documentation of suppliers.	26
• It will be checked that the export of materials for isometric files is used as required, in terms of identifying the line (as line list) and name of the isometric (including number of sheets).	27, 17
• It will be checked that the drawing or isometric format to issue, matches and includes the information required by the project.	

5.- VERIFICATION OF PIPING STRESS AND SUPPORTS

Verification by the specialty will be in two stages, an initial check of the documents and/or calculations of the specialty (see section 5.1) and a second check of testing on the drawings, either in design or supplier drawings, which verifies that the design correspond with the definition, calculation and/or specifications of the specialty that apply to the project.

Unless otherwise stated, in the event that the specialty jobs have been entrusted to partner companies, consultants and external advisors, verification is carried out by the Intecsa Industrial specialist responsible for overseeing the work in question.

5.1. Activity 34 – Material Specifications and MTO

a) Piping Material Specifications (Item 3410)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• It will be checked that specification classes required have been, and that they agree with the Design Basis and regulations to implement to the project. • It will be checked that have been included the particular specifications to complement the general to the specific project in question.	1, 3, 17, 20

b) Material take-off (MTO's) (Item 3420)

Scope of verification	Documents needed for verification(see section 3 of this IT)
<u>Material Take-offs</u>	
• It will be verified the content of the lists of materials in terms of:	
. Correct identification in the header of the data related to measurement (OT number, edition) and the line (area, line number, drawing number, etc.).	1, 3, 7, 17, 21
. The transcript of the codes and modifiers related elements and assemblies and the correct indication of the diameter/s relating to such elements or assemblies.	2, 21
. The reference to the specification class for each elements and ensembles.	2, 3, 17, 21, 22
. The note code indication where appropriate.	3, 21
• Check that all the lines of the list of lines are included in the measurement, in order to ensure (each measurement) has been covered the entire of the facility.	3, 10, 17, 21

c) Isometrics, Tie-in´s and Material Lists (Items 3440/3445)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Check that the lists of materials cover all the elements shown in them.	2, 3, 9,10, 17
• It will verify the content of the lists of materials in terms of:	
. The correct identification of the data for the project and the line (area, unit, line number, drawing number, etc.).	1,10, 17
. The pipes codes transcript and their short description match with the specified for that pipe and its class, in the event that the lists are generated by the specialty materials.	2, 10, 17
. The accuracy of the amounts shown opposite to the representation of line on the drawing.	10
. The reference to the specification class for each of the elements.	2, 3, 10, 17
• The display of notes where appropriate.	2, 3, 10, 17
• It will be controlled that speciality Items (H) that appear designed in isometric are included in the List of H Items and P & ID's	3, 10, 23
• The design shall be in accordance with the standards and material specifications of the project.	2, 3, 10, 17, 23

d) Material requirements (RM's), technical comparisons and vendor documentation (Item 3450)

Scope of verification	Documents needed for verification(see section 3 of this IT)
Material requirements	
• It will be checked that material requirements have been executed in accordance with the Procedure PR-33.	24
Technical comparisons	
• It will be checked that the technical tenders received met as described in the material requirements, specifications and standards	2, 24, 25
Vendor documentation	
• It will be verified that the technical documentation received, complies with the requested on the material requirements.	24, 26

e) Special elements of pipes (H Items) (Item 3430)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• When the Process Department generates Data Sheets it will be verified the correct definition of H Items completing the details of their characteristics according to the design basis and the rules applicable to the project.	1, 2, 3, 10, 17,23
• It will be verified that all special items are included in P & ID's and list of special items.	3, 23

f) Fluid List and Pipe List (Item 3435)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• It will be checked that the correct allocation of piping material classes and the rating regarding the design conditions (design pressure and temperature, overthickness of corrosion and base material) in accordance with the design basis and the rules to be applied to project.	1, 2, 17,20
• Check for a list of lines that classes of piping material specifications assigned are capable of supporting the design conditions for internal and external pressures, as required.	2, 17, 20
• It will be verified check pressure included on the list of lines according to the applicable regulations and design basis of the project.	17

g) European Directive of pressure equipments (PED) (Item 3460)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• In case you require it, we will check the correct assignment of the categories in the list of lines and piping elements, whether bulk material or pipe special elements.	17, 23
• It will be checked that all documentation that is within the scope of the specialty, satisfies the requirements of the standard.	2, 24, 25, 26

h) Calculations of thickness to internal and external pressure (Item 3470)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• It will be checked that the thickness for internal pressure of the piping material classes, are calculated according to the rules applicable to the project.	2, 17, 23, 25, 26
• In the case that a class of piping material specifications will be subject to external pressures, check that the thicknesses of the classes of piping material specifications are calculated according to the standard for external pressure applied to the project.	2, 17, 23

i) Package Units (Items 3497)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Check that in the Material Requisition (MR) is included the standards and specifications that apply. • Verification of documentation will depend on the type of unit package and the type of project.	2, 17, 23, 25, 26

6.- VERIFICATION OF PIPING STRESS AND SUPPORTS

Verification by the specialty will be in two stages, an initial stage of checking of the documents and / or calculations of the specialty (see section 6.1) and a second stage of testing on the drawings, either in design or suppliers, which verifies that the design satisfies the definition, calculation and / or specifications of the specialty that apply to the project.

Unless otherwise stated, in the case that the specialty tasks have been entrusted to partner companies, consultants and external advisors, verification is carried out by the Intecsa Industrial specialist, responsible for overseeing the work in question.

6.1. Activity 35 – Stress and Supports

a) Calculation of stresses in pipes and equipments (Item 3510)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Check that the calculations are made according to the rules applicable at the project and to the Department Job Instructions that apply. Calculations contain the following points: • **Brief description of the item to be calculated.** It will be briefly explained the element in question. . **Scheme of the element to be calculated**. The scheme of the element to be calculated will be accurately represented, its representation if required. . **Calculation hypothesis**. They will be fully in accordance with the applicable Project Basis. . **Determination of applications for each hypothesis separately**. It have been solved the calculation for each scheme of the previous section. They have attached computerized calculation results, in its case. . **Sizing of the elements for the worst hypothesis**. Consist of the material characteristics, the final tension, deformation, etc. • It will be checked on the piping drawings and / or on the model that it has been correctly entered the comments derivatives of piping flexibility analysis calculations performed. In the case of a 3D project it will not be verified with civil planes but is made directly on the 3D model.	1, 2, 3, 5, 7, 11 10, 12, 17

b) Ducts (Item 3511)

Ducts are outside the scope of the Department, (those are the responsibility of the Mechanic Mechanics). In the case that the support of them is requested in the project, the scope of the verification will be:

Scope of verification	Documents needed for verification(see section 3 of this IT)
• It will be checked on the plant drawing or on isometric, as defined for the project, that all ducts are adequately supported, and that the elements for supports, anchors, guides or other types of structural support of design have been properly represented, located and identified.	1, 3, 5, 12, 28
• Check that the inherent notes are properly indicated to the installation of ducts and its elements to be supported as well as the references to relevant documents (standard drawings, plans of special media, etc.).	28

c) Calculation and design of supports (Item 3520)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Check on the plant drawings or on isometric, as defined for the project, that all pipes are adequately supported, and that the elements for supports, anchors, guides or other types of structural supports of design have been properly represented, located and identified.	1, 2, 3, 5, 7 (8), 10, 12, 17
• Check that the inherent notes are properly indicated to the installation of ducts and its elements to be supported as well as the references to relevant documents (standard drawings, plans of special supports, etc.).	7, 10

d) Support List (Item 3530)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• It will be checked that the correct inclusion of all existing supports as well as references of relevant documents (standard drawings, planes of special supports, etc.).	1, 15, 19

e) Special Elements (Item 3540)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Check the correct definition of special items resulting from calculations that are derived from flexibility analysis and detail their characteristics. • It will be checked on the plant drawing or on isometric, as defined for the project, that all ducts are adequately supported, and that the elements for supports, anchors, guides or other types of structural support of design have been properly represented, located and identified.	1, 2, 3, 5, 7 (8), 10, 12, 17

f) European Directive of pressure equipments (PED) (Item 3560)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Check that all documentation included in the dossier complies with the requirements of the standard and that is the last edition.	17

g) Standard Supports in 3D (Item 3596)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• Check location, size, orientation, and possible interference of the supports in structures, pipes, etc.	27, 29,30
• Confirm that the support designed matches with the one included in the 3D model.	27, 29,30
• Verify that all supports are included in the 3D model and are in the correct line.	27, 29,30

h) Package Units (Items 3594)

Scope of verification	Documents needed for verification(see section 3 of this IT)
• It will be verified the possible efforts/supports nearby connection points with pipes designed by Intecsa Industrial.	10, 17, 26

i) Material requirements (RM's), technical comparisons and vendor documentation (Item 3540).

Scope of verification	Documents needed for verification(see section 3 of this IT)
Material requirement . It will be checked that material requirements have been executed in accordance with the Procedure PR-33.	24
Technical comparisons It will be checked that the technical tenders received met as described in the material requirements, specifications and standards.	24, 25
Vendor documentation It will be verified that the technical documentation received, complies with the requested on the material requirements.	24, 26

j) Material take-off (MTO's) of tubular material (Item 3530)

Scope of verification	Documents needed for verification(see section 3 of this IT)
Material take-off • It will be checked for standard support drawings and special supports that sheets of tubular material (pipes and elbows) per line cover all those represented in the same.	27, 29, 30

ANNEXED 1

VERIFICATION PLAN

ABBREVIATIONS FOR VERIFICATION PLANS

PR	Process
CW	Civil works
AR	Architecture
ME	Mechanical
PMS	Piping Materials Specialist
SS	Piping Stress and Supports
PD	Piping Design
FA	Facilities
EL	Electricity
IN	Instrumentation

PM	Project Manager
DM	Department Manager
DL	Discipline leader
SE	Specialist Engineer
TL	Design Team Leader
ADM	Catalog Administrator
DE	Designer
DR	Draftsman

SUBC	Subcontractor
SUPP	Supplier
H	Speciality Items
MR	Material Requisitions
TBA	Technical Bid Analysis

Levels of verification

A	Complete verification by someone other than the author
B	Complete verification of a sample of documents by someone other than the author
C	Verification by the author

ITEM	DESCRIPTION	LEVEL	PROD.	VERIF.	CO-VERIF.	APROV. TEC.
	ACTIVITY 34 –PIPING MATERIALS					
3410	Piping material specifications	C	SE-PMS	SE-PMS	---	DL-PMS
3420	Material take-off, MTO's	C	SE-PMS	SE-PMS	---	DL-PMS
3430	H Items	C	SE-PMS	SE-PMS	---	DL-PMS
3435	Fluid List and Pipe List	C	SE-PMS	SE-PMS		DL-PMS
3440	Material List by isometric	C	SE-PMS	SE-PMS		DL-PMS
3445	TIE-in verification	C	SE-PMS	SE-PMS		DL-PMS
3450	MR s, TBA. Verification .Supplier Documents	C	SE-PMS	SE-PMS		DL-PMS
3460	Calculation/Documentation PED-CE	C	SE-PMS	SE-PMS		DL-PMS
3470	Thickness calculation and vacuum	C	SE-PMS	SE-PMS		DL-PMS
3492	Inspection					
3493	Assisting commissioning, test					
3495	Field data collection					
3496	Construction Support / SEI					
3497	Verification of package units	C	SUPP.	SE-PMS		DL-PMS
3498	Subcontractors Engineering Supervision. Piping Materials	B	SUBC.	SUBC.	SE-PMS	DL-PMS
3499	General Piping Materials					
	ACTIVITY 35 – PIPING STRESS AND SUPPORTS					
3510	Piping and equipment ent stress calculation	C	SE-SS	SE-SS	---	DL-SS
3511	Ducts	C	SE-SS	SE-SS		DL-SS
3520	Supports design and calculation	C	SE-SS	SE-SS	---	DL-SS
3530	Standard supports list	C	SE-SS	SE-SS	---	DL-SS
3540	Special Elements, MR´s	C	SE-SS	SE-SS	----	DL-SS
3560	Calculations/Documentation PED-CE	C	SE-SS	SE-SS		DL-SS
3592	Inspection					
3594	Package units verification	C	SUPP.	SE-SS		DL-SS
3595	Field data collection					
3596	3D support standard	C	SE-SS	SE-SS		DL-SS
3597	Construction Support					
3598	Subcontractors Engineering Supervision. Efforts.	B	SUBC.	SUBC.	SE-SS	DL-SS
3599	General. Stress and Supports.					

ITEM	DESCRIPTION	LEVEL	PROD.	VERIF.	CO-VERIF.	APROV. TEC.
	ACTIVITY 36 – PIPING DESIGN					
2010	Plant process diagram (delineation)	A	DR-PD	PR		DL-PR
2020	Utilities diagram (delineation)	A	DR-PD	PR		DL-PR
2030	Off-sites diagram (delineation)	A	DR-PD	PR		DL-PR
2040	Balance process plant (delineation)	A	DR-PD	PR		DL-PR
2050	Utilities balance (delineation)	A	DR-PD	PR		DL-PR
2060	Off-sites balance (delineation)	A	DR-PD	PR		DL-PR
2110	Process units P&I (delineation)	A	DR-PD	PR/IN		DL-PR
2120	P&I Utilities (delineation)	A	DR-PD	PR/IN		DL-PR
2130	P&I Off-sites (delineation)	A	DR-PD	PR/IN		DL-PR
2140	P&I Distribution (delineation)	A	DR-PD	PR/IN		DL-PR
2150	P&I Interconnections (delineation)	A	DR-PD	PR/IN		DL-PR
2310	Pipe List (Comments)	C	DE-PD	TL-PD	PR	DL-PR
3621	Delineation in CAD of diagrams (PFD´s and P&ID´s)	A	DR-PD	PR		DL-PR
3622	Delineation of intelligent diagrams	A	DR-PD	PR		DL-PR
3623	Delineation in CAD of Plot Plans	A	DR-PD	DL-PD	DL-PR	DL-PD
3624	Delineation in CAD of Piping Drawings	A	DR-PD	TL-PD		DL-PD
3625	Delineation in RASTER of Piping Drawings	A	DR-PD	TL-PD		DL-PD
3626	Delineation of Piping Rack Drawings	A	DR-PD	TL-PD		DL-PD
3627	Delineation Piping Drawings Draft PDMS/PDS	A	DR-PD	TL-PD		DL-PD
3628	Vectorizing Drawings	A	DR-PD	TL-PD		DL-PD
3629	Delineation of disassembly piping drawings	A	DR-PD	TL-PD		DL-PD
3630	General Plot Plan (study)	A	DL-PD/DE-PD			
3631	Key Plan	C	TL-PD/DE-PD	TL-PD		DL-PD
3640	Piping design in electronic model	A	DE-PD	TL-PD		DL-PD
3645	Transformation of electronic model PDMS/PDS	A	DE-PD	DE-PD		DL-PD
3650	Piping plant drawings Design (2D)	A	DE-PD	TL-PD		DL-PD
3651	Study of pipes for model	C	DE-PD	TL-PD		DL-PD
3652	Rack drawings design (2D)	A	DE-PD	TL-PD		DL-PD
3653	Auxiliary drawings, Vents Drains	A	DE-PD	TL-PD		DL-PD
3660	Bounded isometrics	A	DE-PD	TL-PD	SE/ PMS/SS	DL-PD
3661	Non-bounded isometrics	A	DR-PD	TL-PD	SE/ PMS/SS	DL-PD
3662	TIE-in´s	A	DE-PD/DR-PD	TL-PD	SE/PMS	DL-PD
3665	Pipe manual measurement	C	DE-PD			
3668	Verification of package units	A	SUPP	TL-PD		DL PD
3670	Assembly the model	A	DR-PD	TL-PD	SE/PR	DL-PD

ITEM	DESCRIPTION	LEVEL	PROD.	VERIF.	CO-VERIF.	APROV. TEC.
3680	Information for civil	---	TL-PD	---	---	DL-PD
3681	Comments of P&I and Pipe List					
3688	Field data collection					
3692	Inspection					
3693	Assistance to commissioning, testing					
3694	As Built					
3695	Piping design specifications	A	TL-PD	DL-PD	---	DL-PD
3696	3D Administration (PDMS/PDS)	C	ADM-PD	ADM-PD		ADM-PD
3697	2D Administration (I-SKETCH)	C	ADM-PD	ADM-PD		ADM-PD
3698	Engineering Supervision of Subcontractors. Design pipe	B	SUBC.	TL-PD	---	DL-PD
3699	General Piping Design					